

Dr. Yifan Zhou

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Current Position

2023–present **Assistant Professor**
Department of Astronomy, University of Virginia

Education

2014–2019 **Ph.D. in Astronomy and Astrophysics**
Department of Astronomy, University of Arizona
Time-resolved observations of directly-imaged planetary-mass companions and exoplanets.
Advisor: Dániel Apai.

2010–2014 **B.Sc. in Astronomy**
Department of Astronomy, Peking University

Previous Positions

2021–2023 **51 Pegasi b Fellow**
Department of Astronomy, The University of Texas at Austin

2019–2021 **Harlan J. Smith McDonald Observatory Postdoctoral Fellow**
McDonald Observatory / Department of Astronomy, The University of Texas at Austin

Awards and Honors

08/2026 The Schmidt Observatory System FirstLight Awards, co-PI

04/2026 Dean's Research Excellence Fellow, University of Virginia, College of Arts and Sciences

2021–2023 51 Pegasi b Fellowship

2019–2021 Harlan J. Smith McDonald Observatory Fellowship

04/2019 University of Arizona College of Science Excellence in Teaching Award

2015–2018 NASA Earth and Space Science (NESSF) Fellowship

2015–2016 University of Arizona Technology & Research Initiative Funding (TRIF) Fellowship

09/2013 First Lin-Qiao Prize for Excellent Undergraduate Research Projects in Astronomy and Astrophysics, Peking University & KIAA

Grants Awarded

○ 2026 The Schmidt Observatory System FirstLight Awards, Schmidt Sciences (co-PI, amount pending),
Illuminating the Mechanisms of Giant Planet Formation With All Four Schmidt Observatories in Concert

- 2026 James Webb Space Telescope GO-11166, STScI (co-PI, \$ 50k allocated to UVA),
Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with PRISM Phase Curves
- 2026 James Webb Space Telescope GO-11740, STScI (co-PI, \$ 40k allocated to UVA),
A variability survey of temperate brown dwarf atmospheres
- 2026 James Webb Space Telescope GO-11864, STScI (admin-PI, \$ 39k allocated to UVA),
Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c
- 2025 James Webb Space Telescope GO-08846, STScI (co-I, \$ 156k allocated to UVA),
The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwarfs
- 2024 James Webb Space Telescope GO-04967, STScI (PI, \$ 254k),
Dwarfs of Fire and Ice: Mapping the Irradiated Atmospheres of White Dwarf-Brown Dwarf Binaries with NIRSpec PRISM Phase Curves
- 2024 James Webb Space Telescope GO-04758, STScI (PI, \$ 267k),
From Day to Season: Constraining the Rotation Period and Obliquity of Beta Pic b with Time-resolved High-contrast Imaging
- 2023 Heising-Simons Foundation 51 Pegasi b Alumni Faculty Grant (\$ 128k), *Unveiling the Evolution of Planetary Systems*
- 2023 Hubble Space Telescope GO-17427, STScI (PI, \$ 75k),
Monitoring the H-alpha Emission from the PDS 70 planets
- 2023 James Webb Space Telescope GO-03181, STScI (PI, \$ 189k),
Monitor a variable planetary mass companion with NIRSpec IFU
- 2022 Hubble Space Telescope GO-17280, STScI (PI, \$ 87k),
Validating and Characterizing the Protoplanet Candidate AB Aur b with WFC3/UVIS UV and Optical Photometry
- 2021 Hubble Space Telescope GO-16651, STScI (Sci PI, \$ 133k),
A Search for Accreting Protoplanets within Transition Disk Gaps.
- 2023B NASA/Keck Observing Support (PI, \$ 13k)
High-contrast imaging observations of substellar companions of an accelerating star
- 2023A NASA/Keck Observing Support (Sci PI, \$ 13k)
Characterizing the Atmosphere of a Benchmark T Dwarf Companion.
- 2021 NASA/Keck Observing Support (Sci PI, \$ 12k)
The Angular Momentum Architecture of the VHS1256 Planetary System.
- 2019 Hubble Space Telescope GO-16036, STScI (PI, \$ 116k),
Mapping Clouds on a Variable Planetary-Mass Companion.
- 2019 Hubble Space Telescope GO-15830, STScI (Sci PI, \$ 130k),
A Planet is Born: Investigating the Accretion Process of PDS70b with WFC3/UVIS Direct Imaging Observations.
- 2017 Hubble Space Telescope AR-15060, STScI (Sci PI, \$ 150k),
Unleashing the Charges: An Improved Reduction of Key Exoplanet Datasets and a Tool for Ramp Effect Correction.

Selected Observing Programs

As principal investigator or co-principal investigator:

- JWST Cycle 5 (GO-11166, co-PI), *Small But Mighty: Probing Two of the Lowest Mass Survivors of Common Envelope Evolution with PRISM Phase Curves*
- JWST Cycle 5 (GO-11740, co-PI), *A variability survey of temperate brown dwarf atmospheres*
- JWST Cycle 3 (GO-04967), *Dwarfs of Fire and Ice: Mapping the Irradiated Atmospheres of White Dwarf-Brown Dwarf Binaries with NIRSpec PRISM Phase Curves*
- JWST Cycle 3 (GO-04758), *From Day to Season: Constraining the Rotation Period and Obliquity of Beta Pic b with Time-resolved High-contrast Imaging*
- JWST Cycle 3 (GO-06139, co-PI), *Angular Momentum Architecture of the HR 8799 Planetary System*
- JWST Cycle 2 (GO-03181), *Monitor a variable planetary mass companion with NIRSpec IFU*
- JWST Cycle 2 (GO-03375, co-PI), *Dancing 1 - 14 micron spectra to solve the cloudy and chemical puzzle of brown dwarf variability*

- HST Cycle 31 (GO-17427), *Monitoring the H-alpha Emission from the PDS 70 planets*
- HST Cycle 30 (GO-17280), *Validating and Characterizing the Protoplanet Candidate AB Aur b with WFC3/UVIS UV and Optical Photometry*
- HST Cycle 30 (GO-17168, co-PI), *Confirming the Protoplanet Candidate AB Aur b with Accretion Light Echoes*
- HST Cycle 29 (GO-16651), *A Search for Accreting Protoplanets within Transition Disk Gaps*
- HST Cycle 27 (GO-16036), *Mapping Clouds on a Variable Planetary-Mass Companion*
- HST Cycle 27 (GO-15830), *A Planet is Born: Investigating the Accretion Process of PDS70b with WFC3/UVIS Direct Imaging Observations*
- HST Cycle 25 (AR-15060), *Unleashing the Charges: An Improved Reduction of Key Exoplanet Datasets and a Tool for Ramp Effect Correction*
- Spitzer DDT program (14312), *Rotational modulations of a highly variable planetary-mass companion*
- Keck 2023B, *High-contrast imaging observations of substellar companions of an accelerating star*
- Keck 2023A, *Characterizing the Atmosphere of a Benchmark T Dwarf Companion*
- Keck 2021A, *The Angular Momentum Architecture of the VHS1256 Planetary System*
- McDonald Observatory 2022-1, *Monitoring variable brown dwarfs with DIAFI*
- McDonald Observatory 2021-3, *DIAFI z-band monitoring of four highly variable brown dwarfs*

As co-investigator:

- JWST Cycle 5 (GO-11864), *Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c*
- JWST Cycle 4 (GO-08846, PI: N. Fuda), *The Polar Expedition: Extracting Spectra of Polar Regions in Planetary-mass Objects and Brown Dwarfs*

- JWST Cycle 4 (AR-08604, PI: P. Arras), *Debris Disks as the Origin of High Altitude Clouds in Exoplanet Atmospheres*
- JWST Cycle 4 (GO-08581, PI: R. Bowens-Rubin), *The HOTH Survey: Hot On The Hunt for frigid exoplanets in the solar neighborhood*
- JWST Cycle 4 (GO-07612, PI: L. Pearce), *We can directly image super-Earth-sized planets near the habitable zone of Sirius B with JWST/MIRI*
- JWST Cycle 3 (GO-05437, PI: W. Best), *A Census to the Bottom of the IMF in W3: Atmospheres, Disks, Accretion, and Demographics*
- JWST Cycle 3 (GO-05261, PI: H. Jiang), *Confirming the youngest gap-opening protoplanet undergoing runaway gas accretion*
- JWST Cycle 3 (GO-05226, PI: E. Manjavacas), *The Weather Forecast in a Cloudy (or not) Cool Planetary-Mass Brown Dwarf*
- JWST Cycle 2 (GO-03947, PI: B. Bowler), *Testing the Giant Planet Hypothesis for Spiral-driven Arms in Protoplanetary Disks*
- JWST Cycle 2 (GO-02965, PI: B. Biller), *Clouds or Chemistry?: Pinpointing the drivers of variability across the L/T transition via the benchmark L/T binary WISE 1049AB*
- JWST Cycle 1 (GO-02640, PI: W. Best), *A Census to the Bottom of the IMF in Westerlund 2: Atmospheres, Disks, Accretion, and Demographics*
- JWST Cycle 1 (GO-02311, PI: Y. Wu/B. Bowler), *JWST MIRI Imaging Survey of Planetary-mass Companions: Testing the Compact Disk Hypothesis*
- JWST Early Release Science Program (ERS-1386, PI: S. Hinkley), *High Contrast Imaging of Exoplanets and Exoplanetary Systems with JWST*
- HST Cycle 34 (GO-18376, PI: Crotts), *Extending Accretion Studies into the Sub-Jupiter Regime with TWA 7 b*
- HST Cycle 34 (GO-18357, PI: Sun), *Domain expansion: Detecting the first extrasolar UV aurora with deep COS observations*
- HST Cycle 33 (GO-18139, PI: L. Jiang), *Tracing Accretion in the Planetary Regime: A Comprehensive UV/Optical Survey of the Late Stages of Planet Formation*
- HST Cycle 33 (GO-18114, PI: G. Cugno), *Unveiling Hidden Protoplanets: Direct Extinction Measurement at Optical Wavelengths in the Planet-Formed Gap of AS209*
- HST Cycle 33 (GO-18113, PI: C. O'Toole), *UV Exo-Aurorae: Multi-Wavelength Observations of a Radio Emitting T-dwarf to confirm Auroral Activity in the Ultraviolet*
- HST Cycle 32 (GO-17768, PI: L. Flagg), *Detecting the UV Accretion Signatures from PDS-70b*
- HST Cycle 31 (GO-17283, PI: G. Cugno), *Confirming an accreting planet in a very young protoplanetary disk*
- HST Cycle 30 (GO-17136, PI: E. Gaidos), *Photometry of a Young Planetary-Mass Companion to a Taurus M Dwarf Star*
- HST Cycle 30 (GO-17127, PI: Y. Aoyama), *Testing models of accretion onto the Young Planetary System PDS 70*
- HST Cycle 29 (GO-16754, PI: S. Casewell), *Decoding the clouds on an irradiated inflated brown dwarf*
- HST Cycle 29 (GO-16721, PI: B. Bowler), *The Angular Momentum Architecture of Long-Period Giant Planets and Brown Dwarf Companions*

- HST Cycle 28 (GO-16302, PI: Y. Wu), *Accretion Rates as a Diagnostic Tool for the Origin of Planetary-mass Companions*
- HST Cycle 27 (GO-15947, PI: D. Apai), *Dancing with the Dwarfs: Very High Quality Spatial and Spectral Maps of Hot Jupiters Proxies*
- HST Cycle 25 (GO-15301, PI: L. Carone), *Now You See Me – the WASP-117b Version*
- HST Cycle 23 (GO-14241, PI: D. Apai), *Cloud Atlas: Vertical Cloud Structure and Gravity in Exoplanet and Brown Dwarf Atmospheres*

Service and Committees

- UVA colloquium committee member; AURA representative.
- Referee for the AAS Journals (AJ, ApJ, ApJL), A&A, MNRAS, Nature Astronomy, and Nature; 3–5 manuscripts per year.
- NSF Astronomy and Astrophysics Grants panel reviewer
- NASA Exoplanet Research Program panel reviewer
- Canadian Time Allocation (CanTAC) Committee 2020–2023
- 2021 UT Austin Department of Astronomy graduate admissions committee
- James Webb Space Telescope Proposal Panelist (Cycles 2, 3, 4)
- Hubble Space Telescope Proposal Reviewer and Panelist (Cycles 27, 29, 30, 34)
- 2020 NASA FINESST Fellowship Reviewer
- 2018 University of Arizona Department of Astronomy graduate admissions committee

Teaching Activities

- Mentor to UVA undergraduate astronomy minors.
- UVA ASTR-5110 Astronomical Techniques: Fall 2025.
- UVA ASTR-1250 Alien Worlds: Spring and Fall 2024; Spring 2025; Spring 2026.
- UVA ASTR-1610 Intro to Astronomical Research: Spring 2024; Spring 2025; one guest lecture per semester.
- UVA College Science Scholars Presentation: Spring 2025; Spring 2026; one 75-minute lecture.
- University of Arizona Teaching Assistant for Dr. Laird Close, course name: *Life in the Universe*
Gave three 50-minute lectures.
- University of Arizona Teaching Assistant for Dr. Serena Kim, course name: *Life in the Universe*
Organized in-class activities; Gave two 75-minute lectures.

Mentoring

- Current Postdocs: Arthur Adams (2024–present), Dorian Demars (2025–present)
- UVA graduate students: Daphne Broski-Laing (2024–present), Danya Alboslani (2025–present)

- Undergraduate Researchers:
 - UVA VICO summer research students:
Yihan Li (Peking University, 2025–2026), Emma Pavlicek (University of Chicago, 2025–2026), Shubhrangshu Debsarkar (UVA, 2025), Aryaa Patel (UVA, 2025–2026), Benjamin Chang (UVA, 2024–2026), Maddie Chapleski (UVA, 2024–2026), Sawyer Tabor (UVA, 2023–2025)
 - UVA senior thesis and undergraduate research assistants:
Casey York (UVA, 2026–present), Eleanor Press (UVA, 2026–present) Jumi Hall (UVA, 2023–2024), Malhar Kulkarni (UVA, 2024), Cassidy Walker (UVA, 2024–2025), Charlene Huynh (UVA, 2025–2026), Anna You (UVA, 2025–present)
 - UT Austin TAURUS scholars:
Mateo Guerra Toro (2021–2022), Destiny Howell (2022–2023)
 - UT Austin undergraduate research assistants:
Aniket Sanghi (2021–2023), Neel Nagarajan (2023–2024)
 - University of Arizona undergraduate research assistants: Zhanbo Zhang (2018)

Invited Talks and Colloquia

- 04/2026 Invited Seminar, Department of Astronomy, Princeton University, Princeton, NJ
- 03/2026 Invited Seminar, Department of Astronomy, Johns Hopkins University, Baltimore, MD
- 03/2026 Invited Seminar, Department of Earth and Atmospheric Physics, Massachusetts Institute of Technology, Cambridge, MA
- 03/2026 Colloquium, Department of Physics, Amherst College, Amherst, MA
- 02/2026 Colloquium, Department of Astronomy, The Pennsylvania State University, State College, PA
- 12/2025 Colloquium, Department of Astronomy, Zhejiang University, Hangzhou, China
- 12/2025 Invited talk, Exoplanets and Planet Formation, Shanghai, China
- 06/2025 Invited talk, Kavli Summer Program in Astrophysics, Charlottesville, VA
- 05/2025 Astronomy Seminar, American Museum of Natural History, New York, NY
- 05/2025 Astronomy Seminar, Center for Computational Astrophysics, New York, NY
- 03/2025 MPIA Exo-coffee, Heidelberg, Germany
- 03/2025 Review Talk, Gas Accretion in Planet Formation Conference, Heidelberg, Germany
- 11/2024 Colloquium, Department of Astronomy, University of Wisconsin–Madison, Madison, WI
- 08/2024 Lecture, Protoplanetary Disk and Planet Formation Summer School, China Center of Advanced Science and Technology, Beijing, China
- 06/2024 Carnegie EPL Seminar, Carnegie Institute, Washington, DC
- 03/2024 Planetary Science Seminar, Caltech Department of Planetary Science, Pasadena, CA
- 11/2023 Seminar, TD Lee Institute, Shanghai Jiaotong University, China
- 09/2023 Colloquium, Herzberg Astronomy & Astrophysics, National Research Council of Canada
- 02/2023 Colloquium, University of Virginia, Charlottesville, VA
- 02/2023 SPF Seminar, University of Michigan, Ann Arbor, MI
- 01/2023 Origins Seminar, University of Arizona, Tucson, AZ

12/2022 Caltech Tea talk, Caltech, Pasadena, CA
 06/2021 MPIA Exo-coffee (online)
 04/2021 UC Berkeley Planetary Science Seminar, UC Berkeley (online)
 02/2021 Carnegie EPL Seminar, Carnegie Institute (online)
 02/2021 Origins Seminar, University of Arizona (online)
 10/2020 MIT Exoplanet Tea, MIT (online)
 10/2020 AMNH Astrophysics Seminar, American Museum of Natural History (online)
 11/2018 Seminar talk, Jet Propulsion Laboratory, Pasadena, CA, USA
 11/2018 Seminar talk, University of Texas, Austin, TX, USA
 09/2018 Seminar talk, IPAC lunch talk, Pasadena, CA, USA
 02/2018 Seminar talk, CPSX Lunch Forum at Western University, London, ON, Canada

Other Talks and Seminars

07/2026 Contributed talk, Charting Substellar Worlds, Asilomar Conference Grounds, CA, USA
 07/2026 Contributed talk, Exoplanets 6, Porto, Portugal
 07/2025 Contributed talk, Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Washington, DC, USA
 07/2025 Contributed talk, Exoclime VI, Montreal, Canada
 11/2022 SPF Seminar, UT Austin, USA
 08/2022 51 Pegasi b Fellowship Summit, San Francisco, CA, USA
 07/2022 UCSC OWL seminar, Santa Cruz, CA, USA
 05/2022 Contributed talk, Exoplanets IV, Las Vegas, NV, USA
 01/2022 SPF Seminar, UT Austin
 09/2021 Twinkle conference (online)
 08/2021 51 Pegasi b Fellowship Summit (online)
 12/2020 KIAA Lunch Talk, Peking University (online)
 11/2020 SPF Seminar, UT Austin (online)
 02/2020 Seminar talk, University of Texas, Austin, TX, USA
 09/2018 Seminar talk, University of Bern, Bern, Switzerland
 07/2018 Contributed talk, Exoplanet II, Cambridge, UK
 04/2018 Seminar talk, Origins Seminar, Tucson, AZ, USA
 03/2018 Contributed talk, Star and Planet Formation in the Southwest II, Oracle, AZ, USA
 01/2018 Seminar talk, KIAA Lunch Talk, Beijing, China
 12/2017 Seminar talk, PSF coffee, MPIA, Heidelberg, Germany
 10/2017 Contributed talk, BDEXOCON conference, Newark, DE, USA
 05/2017 Seminar talk, Origins Seminar, Tucson, AZ, USA
 08/2016 Contributed talk, Exoclime II, Squamish, Canada

Publications

 [ADS](#),  [Google Scholar](#).

14 published first-author papers. 6 Mentee-led papers. 66 total refereed publications.

Total citations: 2796. *h*-index: 28

First-author papers:

1. Yifan Zhou, Beth A. Biller, Aarynn L. Carter, et al.
Photometric Variability and Rotation of β Pictoris b from JWST NIRCcam Coronagraphic Imaging. In: AJ 172.3, 152 (Sept. 2026), p. 152. [\[ADS\]](#)
2. Yifan Zhou, Brendan P. Bowler, Aniket Sanghi, et al.
Evidence for Variable Accretion onto PDS 70 c and Implications for Protoplanet Detections. In: ApJL 980.2, L39 (Feb. 2025), p. L39. [\[ADS\]](#)
3. Yifan Zhou, Brendan P. Bowler, Haifeng Yang, et al.
UV-optical Emission of AB Aur b Is Consistent with Scattered Stellar Light. In: AJ 166.6, 220 (Dec. 2023), p. 220. [\[ADS\]](#)
4. Yifan Zhou, Brendan P. Bowler, Dániel Apai, et al.
Roaring Storms in the Planetary-mass Companion VHS 1256-1257 b: Hubble Space Telescope Multiepoch Monitoring Reveals Vigorous Evolution in an Ultracool Atmosphere. In: AJ 164.6, 239 (Dec. 2022), p. 239. [\[ADS\]](#)
5. Yifan Zhou, Aniket Sanghi, Brendan P. Bowler, et al.
HST/WFC3 $H\alpha$ Direct-imaging Detection of a Pointlike Source in the Disk Cavity of AB Aur. In: ApJL 934.1, L13 (July 2022), p. L13. [\[ADS\]](#)
6. Yifan Zhou, Dániel Apai, Xianyu Tan, et al.
HST/WFC3 Complete Phase-resolved Spectroscopy of White-dwarf-brown-dwarf Binaries WD 0137 and EPIC 2122. In: AJ 163.1, 17 (Jan. 2022), p. 17. [\[ADS\]](#)
7. Yifan Zhou, Brendan P. Bowler, Kevin R. Wagner, et al.
Hubble Space Telescope UV and $H\alpha$ Measurements of the Accretion Excess Emission from the Young Giant Planet PDS 70 b. In: AJ 161.5, 244 (May 2021), p. 244. [\[ADS\]](#)
8. Yifan Zhou, Brendan P. Bowler, Caroline V. Morley, et al.
Spectral Variability of VHS J1256-1257b from 1 to 5 μ m. In: AJ 160.2, 77 (Aug. 2020), p. 77. [\[ADS\]](#)
9. Yifan Zhou, Dániel Apai, Luigi R. Bedin, et al.
Cloud Atlas: High-precision HST/WFC3/IR Time-resolved Observations of Directly Imaged Exoplanet HD 106906b. In: AJ 159.4, 140 (Apr. 2020), p. 140. [\[ADS\]](#)
10. Yifan Zhou, Dániel Apai, Ben W. P. Lew, et al.
Cloud Atlas: High-contrast Time-resolved Observations of Planetary-mass Companions. In: AJ 157.3, 128 (Mar. 2019), p. 128. [\[ADS\]](#)
11. Yifan Zhou, Dániel Apai, Stanimir Metchev, et al.
Cloud Atlas: Rotational Modulations in the L/T Transition Brown Dwarf Companion HN Peg B. In: AJ 155.3, 132 (Mar. 2018), p. 132. [\[ADS\]](#)

12. Yifan Zhou, Dániel Apai, Ben W. P. Lew, et al.
A Physical Model-based Correction for Charge Traps in the Hubble Space Telescope's Wide Field Camera 3 Near-IR Detector and Its Applications to Transiting Exoplanets and Brown Dwarfs. In: AJ 153.6, 243 (June 2017), p. 243. [ADS]
13. Yifan Zhou, Dániel Apai, Glenn H. Schneider, et al.
Discovery of Rotational Modulations in the Planetary-mass Companion 2M1207b: Intermediate Rotation Period and Heterogeneous Clouds in a Low Gravity Atmosphere. In: ApJ 818.2, 176 (Feb. 2016), p. 176. [ADS]
14. Yifan Zhou, Gregory J. Herczeg, Adam L. Kraus, et al.
Accretion onto Planetary Mass Companions of Low-mass Young Stars. In: ApJL 783.1, L17 (Mar. 2014), p. L17. [ADS]

Mentee-led papers:

15. Daphne Broski-Laing, Yifan Zhou, Joshua D. Lothringer, et al.
Asymmetric nightside CO₂ features, inefficient heat transport, and precise evolutionary constraints: Spectroscopic phase curves reveal the past and present of a white dwarf-brown dwarf binary. In: arXiv e-prints, arXiv:2606.30112 (June 2026), arXiv:2606.30112. [ADS]
16. Yihan Li, Yifan Zhou, Rachel Bowens-Rubin, et al.
A Search for Wide-orbit Planets around M Dwarfs Using Deep MIRI 15 μ m Images. In: AJ 172.1, 26 (July 2026), p. 26. [ADS]
17. Madalyn F. Chapleski and Yifan Zhou.
Gray Spectral Variability in Three Brown Dwarfs Observed by HST/WFC3 Time-series Observations. In: ApJ 997.2, 152 (Feb. 2026), p. 152. [ADS]
18. Arthur D. Adams, Yifan Zhou, Gabriel-Dominique Marleau, et al.
Characterizing the Time Variability of 2M1207 A + b with JWST NIRSpec/PRISM. in: AJ 170.5, 289 (Nov. 2025), p. 289. [ADS]
19. Aniket Sanghi, Yifan Zhou, and Brendan P. Bowler.
Efficiently Imaging Accreting Protoplanets from Space: Reference Star Differential Imaging of the PDS 70 Planetary System Using the HST/WFC3 Archival PSF Library. In: AJ 163.3, 119 (Mar. 2022), p. 119. [ADS]
20. Zhanbo Zhang, Yifan Zhou, Benjamin V. Rackham, et al.
The Near-infrared Transmission Spectra of TRAPPIST-1 Planets b, c, d, e, f, and g and Stellar Contamination in Multi-epoch Transit Spectra. In: AJ 156.4, 178 (Oct. 2018), p. 178. [ADS]

Co-author papers:

21. Merle A. Schrader, Johanna M. Vos, Evert Nasedkin, et al.
The JWST weather report: Unravelling the atmospheric variability of isolated worlds using principal component analysis. In: arXiv e-prints, arXiv:2607.26182 (July 2026), arXiv:2607.26182. [ADS]
22. J. de Wit, R. Alonso, B. V. Rackham, et al.
J0404+1112: A 3-Hour Eclipsing White Dwarf-Brown Dwarf Probing Multiple Atmospheric Regimes. In: arXiv e-prints, arXiv:2606.03737 (June 2026), arXiv:2606.03737. [ADS]

23. Claire O. Finley, Brendan P. Bowler, Ya-Lin Wu, et al.
Ultraviolet Imaging of SR 12 c with HST/WFC3: Accretion and Variability of a Giant Planet at the End Stages of Growth. In: AJ 172.2, 134 (Aug. 2026), p. 134. [ADS]
24. Ben J. Sutcliffe, Markus J. Bonse, Valentin Christiaens, et al.
Direct Imaging Discovery of Giant Exoplanet β Pictoris d: A Decade-long Game of Hide-and-seek. In: ApJL 1006.1, L10 (July 2026), p. L10. [ADS]
25. Xianyu Tan, Xi Zhang, Mark S. Marley, et al.
Large-amplitude variability driven by giant dust storms on a planetary-mass companion. In: Science Advances 11 (Nov. 2025), p. 22.3324. [ADS]
26. J. J. Zanazzi, Eugene Chiang, and Yifan Zhou.
Seismic Oscillations Excited by Giant Impacts in Directly Imaged Giant Planets. In: ApJ 993.1, 3 (Nov. 2025), p. 3. [ADS]
27. Andrew Householder, Mary Anne Limbach, Beth Biller, et al.
Sensitivity to Sub-10-sized Exosatellite Transits in the MIRI LRS Light Curve of the Nearest Substellar Worlds. In: ApJL 992.1, L16 (Oct. 2025), p. L16. [ADS]
28. Brendan P. Bowler, Yifan Zhou, Lauren I. Biddle, et al.
H α Variability of AB Aur b with the Hubble Space Telescope: Probing the Nature of a Protoplanet Candidate with Accretion Light Echoes. In: AJ 169.5, 258 (May 2025), p. 258. [ADS]
29. Shrishmoy Ray, Steph Sallum, Sasha Hinkley, et al.
The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8 μ m. In: ApJL 983.1, L25 (Apr. 2025), p. L25. [ADS]
30. Rachael C. Amaro, Dániel Apai, Yifan Zhou, et al.
Phase-resolved Hubble Space Telescope WFC3 Spectroscopy of the Weakly Irradiated Brown Dwarf GD 1400 and Energy Redistribution–Irradiation Trends in Six White Dwarf–Brown Dwarf Binaries. In: ApJ 979.2, 231 (Feb. 2025), p. 231. [ADS]
31. Allison M. McCarthy, Johanna M. Vos, Philip S. Muirhead, et al.
The JWST Weather Report from the Isolated Exoplanet Analog SIMP 0136+0933: Pressure-dependent Variability Driven by Multiple Mechanisms. In: ApJL 981.2, L22 (Mar. 2025), p. L22. [ADS]
32. Quanyi Liu, Wei Zhu, Yifan Zhou, et al.
Detecting Planetary Oblateness in the Era of JWST: A Case Study of Kepler-167e. In: AJ 169.2, 79 (Feb. 2025), p. 79. [ADS]
33. Michael Poon, Marta L. Bryan, Hanno Rein, et al.
Leaning Sideways: VHS 1256–1257 b is a Super-Jupiter with a Uranus-like Obliquity. In: AJ 168.6, 270 (Dec. 2024), p. 270. [ADS]
34. Jenni R. French, Sarah L. Casewell, Rachael C. Amaro, et al.
The only inflated brown dwarf in an eclipsing white dwarf-brown dwarf binary: WD1032+011B. In: MNRAS 534.3 (Nov. 2024), pp. 2244–2262. [ADS]
35. Kyle Franson, William O. Balmer, Brendan P. Bowler, et al.
JWST/NIRCam 4–5 μ m Imaging of the Giant Planet AF Lep b. In: ApJL 974.1, L11 (Oct. 2024), p. L11. [ADS]

36. Anna Lueber, Kevin Heng, Brendan P. Bowler, et al.
Retrieved atmospheric properties of the sub-stellar object VHS 1256 b with HST, VLT, and JWST spectra.
In: A&A 690, A357 (Oct. 2024), A357. [ADS]
37. Beth A. Biller, Johanna M. Vos, Yifan Zhou, et al.
The JWST weather report from the nearest brown dwarfs I: multiperiod JWST NIRSpec + MIRI monitoring of the benchmark binary brown dwarf WISE 1049AB. in: MNRAS 532.2 (Aug. 2024), pp. 2207–2233.
[ADS]
38. Joshua D. Lothringer, Yifan Zhou, Dániel Apai, et al.
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